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Visualizing and Debugging Your Knowledge Graph

The following step-by-step guide walks through the process to visualize a knowledge graph after it's been constructed by graphrag. Note that some of the settings recommended below are based on our own experience of what works well. Feel free to change and explore other settings for a better visualization experience!

1. Run the Pipeline

Before building an index, please review your `settings.yaml` configuration file and ensure that graphml snapshots is enabled.

```
snapshots:  
  graphml: true
```

After running the indexing pipeline over your data, there will be an output folder (defined by the `storage.base_dir` setting).

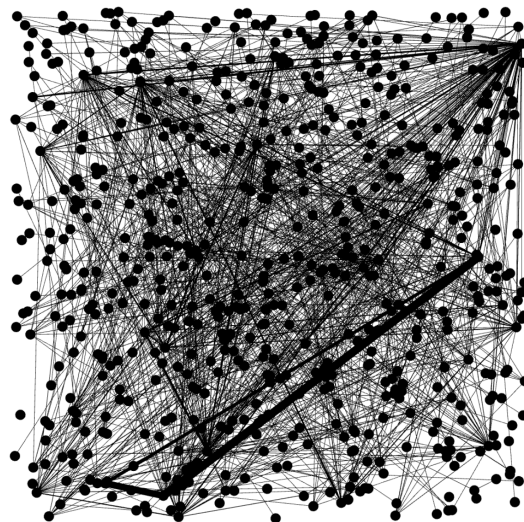
- **Output Folder:** Contains artifacts from the LLM's indexing pass.

2. Locate the Knowledge Graph

In the output folder, look for a file named `graph.graphml`. graphml is a standard [file format](#) supported by many visualization tools. We recommend trying [Gephi](#).

3. Open the Graph in Gephi

1. Install and open Gephi
2. Navigate to the `output` folder containing the various parquet files.
3. Import the `graph.graphml` file into Gephi. This will result in a fairly plain view of the undirected graph nodes and edges.

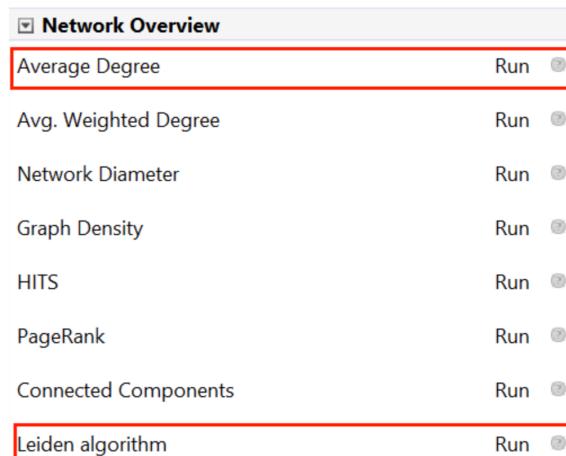


4. Install the Leiden Algorithm Plugin

1. Go to **Tools** -> **Plugins**.
2. Search for "Leiden Algorithm".
3. Click **Install** and restart Gephi.

5. Run Statistics

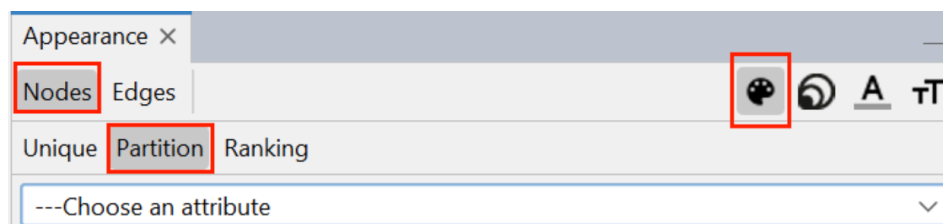
1. In the **Statistics** tab on the right, click **Run** for **Average Degree** and **Leiden Algorithm**.



1. For the Leiden Algorithm, adjust the settings:
2. **Quality function:** Modularity
3. **Resolution:** 1

6. Color the Graph by Clusters

1. Go to the **Appearance** pane in the upper left side of Gephi.



1. Select **Nodes**, then **Partition**, and click the color palette icon in the upper right.
2. Choose **Cluster** from the dropdown.
3. Click the **Palette...** hyperlink, then **Generate...**
4. Uncheck **Limit number of colors**, click **Generate**, and then **Ok**.

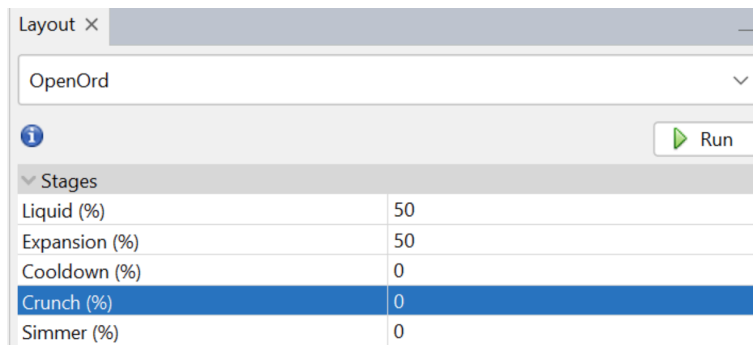
5. Click **Apply** to color the graph. This will color the graph based on the partitions discovered by Leiden.

7. Resize Nodes by Degree Centrality

1. In the **Appearance** pane in the upper left, select **Nodes** -> **Ranking**
2. Select the **Sizing** icon in the upper right.
3. Choose **Degree** and set:
4. **Min:** 10
5. **Max:** 150
6. Click **Apply**.

8. Layout the Graph

1. In the **Layout** tab in the lower left, select **OpenORD**.



1. Set **Liquid** and **Expansion** stages to 50, and everything else to 0.
2. Click **Run** and monitor the progress.

9. Run ForceAtlas2

1. Select **Force Atlas 2** in the layout options.

ForceAtlas 2

Run

Threads	
Threads number	7
Performance	
Tolerance (speed)	1.0
Approximate Repulsion	☐
Approximation	1.2
Tuning	
Scaling	15
Stronger Gravity	☐
Gravity	1.0
Behavior Alternatives	
Dissuade Hubs	☒
LinLog mode	☐
Prevent Overlap	☒
Edge Weight Influence	1.0
Normalize edge weights	☐
Inverted edge weights	☐

1. Adjust the settings:
2. **Scaling**: 15
3. **Dissuade Hubs**: checked
4. **LinLog mode**: uncheck
5. **Prevent Overlap**: checked
6. Click **Run** and wait.
7. Press **Stop** when it looks like the graph nodes have settled and no longer change position significantly.

10. Add Text Labels (Optional)

1. Turn on text labels in the appropriate section.
2. Configure and resize them as needed.

Your final graph should now be visually organized and ready for analysis!